

UNIT 04

STATE OF MATTER (I): GAS



KINETIC MOLECULAR THEORY OF GASES

The experimental observations about the behavior of gases discussed so far can be explained with a simple theoretical model known as the kinetic molecular theory. This theory is based on the following postulates, or assumptions

1. Particles:

A gas is basically a bunch of tiny particles – either atoms or molecules.

2. Teeny Tiny or Point Masses:

These particles making up the gas are super small, like little dots on paper, and they don't weigh much.

3. Negligible volume:

These particles are much smaller than the distance between particles. Most of the volume of a gas is therefore empty space.

4. Nil Force of Interaction:

There is no force of attraction between gas particles or between the particles and the walls of the container.

5. Particles in Motion:

These particles move in a straight line until they collide with another particle or the walls of the container.

6. Container Volume = Gas Volume:

Because these particles are always moving around, they fill up the entire volume of the container. So, just think of the container's volume as the gas volume.

7. Mean free path:

The mean free path is just the average distance a particle travels before bumping into another particle.

8. Kinetic Energy of the particles:

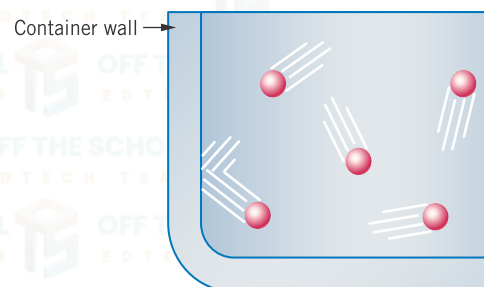
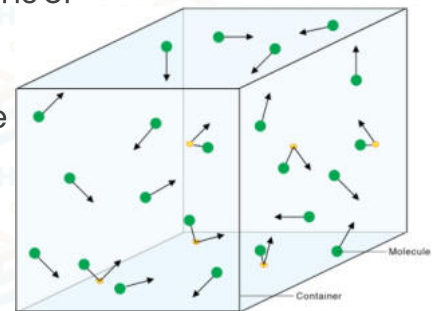
The average kinetic energy of a collection of gas particles depends on the temperature of the gas and nothing else.

9. No Energy or Momentum Makeover:

Collisions between gas particles or collisions with the walls of the container are perfectly elastic. None of the energy of a gas particle is lost when it collides with another particle or with the walls of the container.

10. Pressure of Gas:

When particles hit the container walls, they push with a force. The pressure of the gas is basically how many particles are hitting the walls in a given time per unit area. It's like a collision frequency thing.



Gas molecule = 

 **Gases** Gas molecules are in rapid, random motion.

Properties of Gases

- Volume, V: measured in liters (L)
 - Amount, n: measured in moles
 - Temperature, T: measures in Kelvin (K)
- $K = C + 273.15$

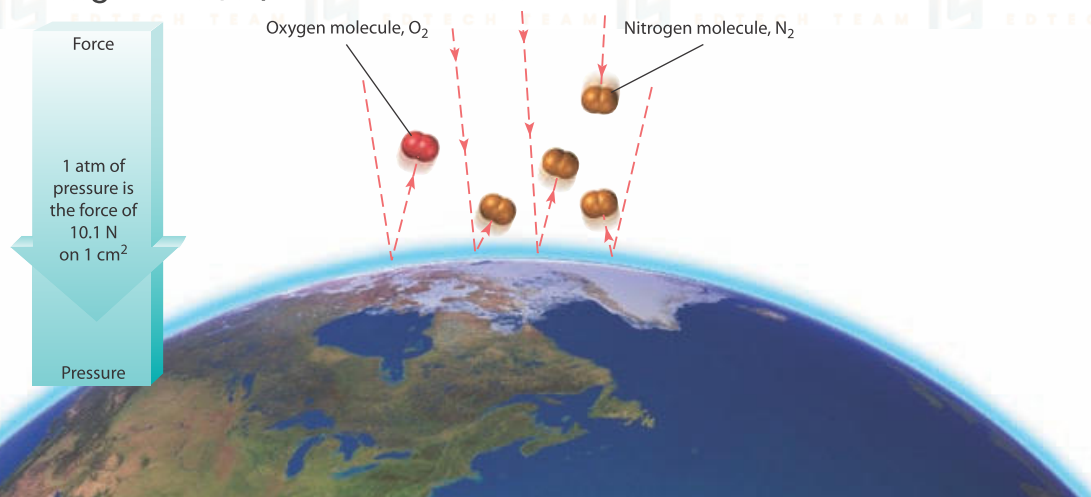
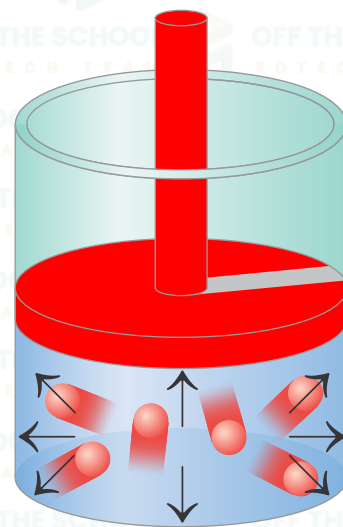


Pressure in a closed system

- $P = \text{force/area}$
- In a closed system, the force is the molecules hitting the sidewall of the container.
- The area is the area of the walls of the container.

Pressure in an open system

- Pressure = force/area.
- The area is the areas of the top of the book.
- Imagine a column of gas extending from your book to the top of the stratosphere. The force comes from the weight of that gas pushing down.
($F = mg$ where $g = 9.8 \text{ m/s}^2$).

**PRESSURE:**

"Force per unit area is pressure."

$$P = \frac{F}{A}$$

UnITS OF PreSSure			
Unit	Symbol	Definition/relationship	Application
pascal	Pa	SI pressure unit $1 \text{ Pa} = \frac{1 \text{ N}}{\text{m}^2}$	scientific (kPa)
millimeter of mercury	mm Hg	pressure that supports a 1 mm mercury column in a barometer	blood pressure monitors
torr	torr	$1 \text{ torr} = 1 \text{ mm Hg}$	vacuum pumps
atmosphere	atm	average atmospheric pressure at sea level and 0°C $1 \text{ atm} = 760 \text{ mm Hg}$ $= 760 \text{ torr}$ $= 1.01325 \times 10^5 \text{ Pa}$ $= 101.325 \text{ kPa}$	atmospheric pressure
pounds per square inch	psi	$1 \text{ psi} = 6.89286 \times 10^3 \text{ Pa}$ $1 \text{ atm} = 14.700 \text{ psi}$	tire gauges

DENSITY:

Mass per unit volume is density or actually it tells the degree of compactness of any matter.

GAS LAWS**PRESSURE-VOLUME LAW (BOYLE'S LAW):****Introduction:**

The relationship between the volume and pressure of a sample of gas under constant-temperature conditions was first described in 1660 by the English scientist Robert Boyle.



Statement:

"The volume of a given mass of a gas is inversely proportional to its applied pressure if the temperature is held constant."

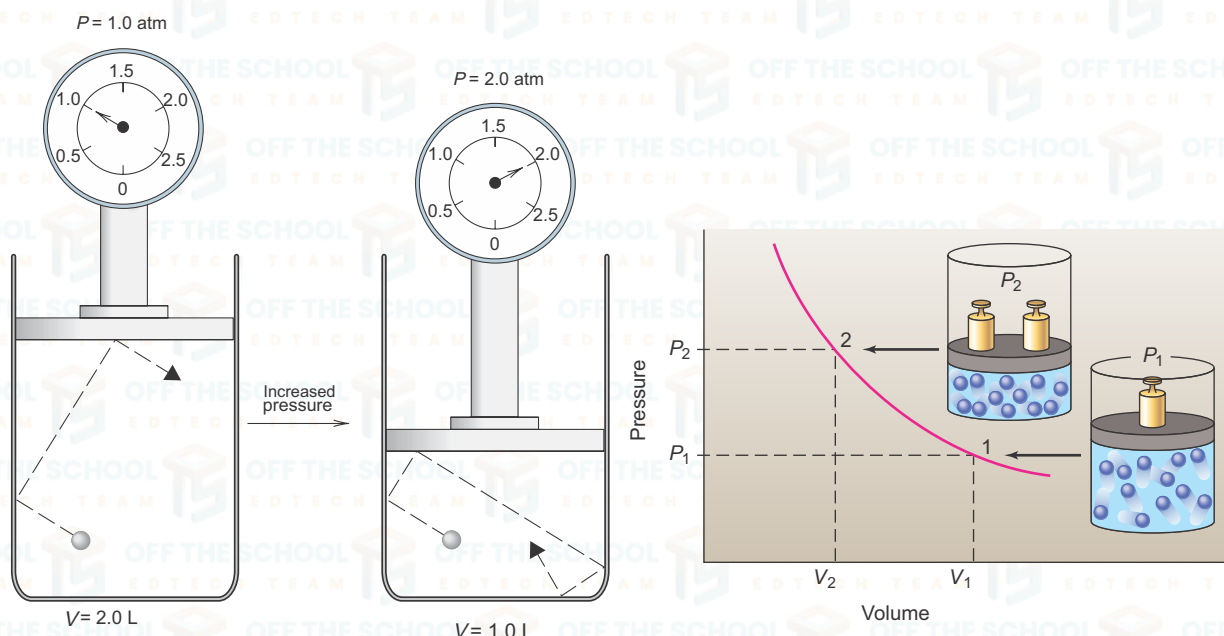
Explanation and derivation:

The inverse relationship suggested by Boyle is represented as;

$$V \propto \frac{1}{P}$$

Or as an equality with k the constant of proportionality:

$$V = k/P \text{ or } PV = k$$

**TEMPERATURE-VOLUME LAW (CHARLES LAW):****Introduction:**

In 1787, Jacques Charles, a balloonist as well as a physicist, proposed that the volume of a gas is related to the temperature. This proposal became Charles's law.

Statement:

"The volume (V) of a gas is directly related to the temperature (T) when there is no change in the pressure (P) and amount (n) of gas."

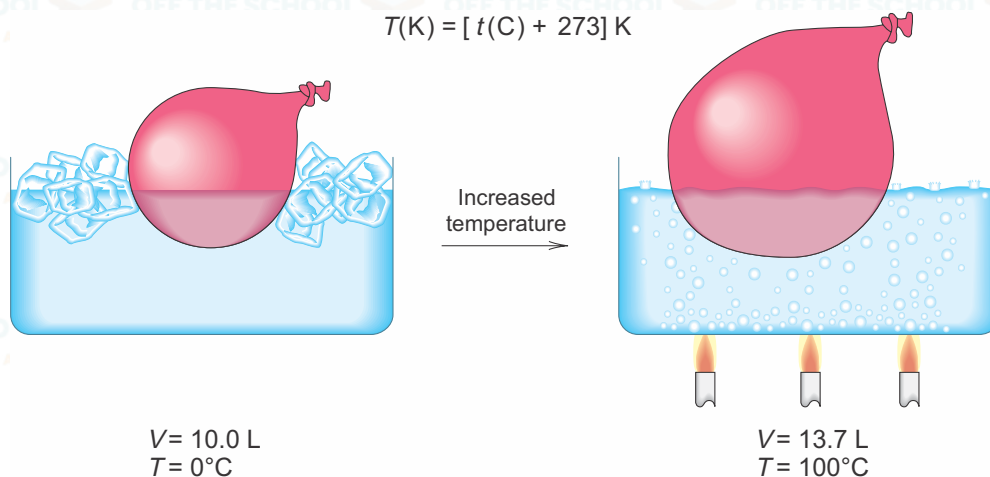
Derivation:

$$V \propto T$$

$$V = k T \text{ or } \frac{V}{T} = k$$

For a quantity of gas at two temperatures at the same pressure, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

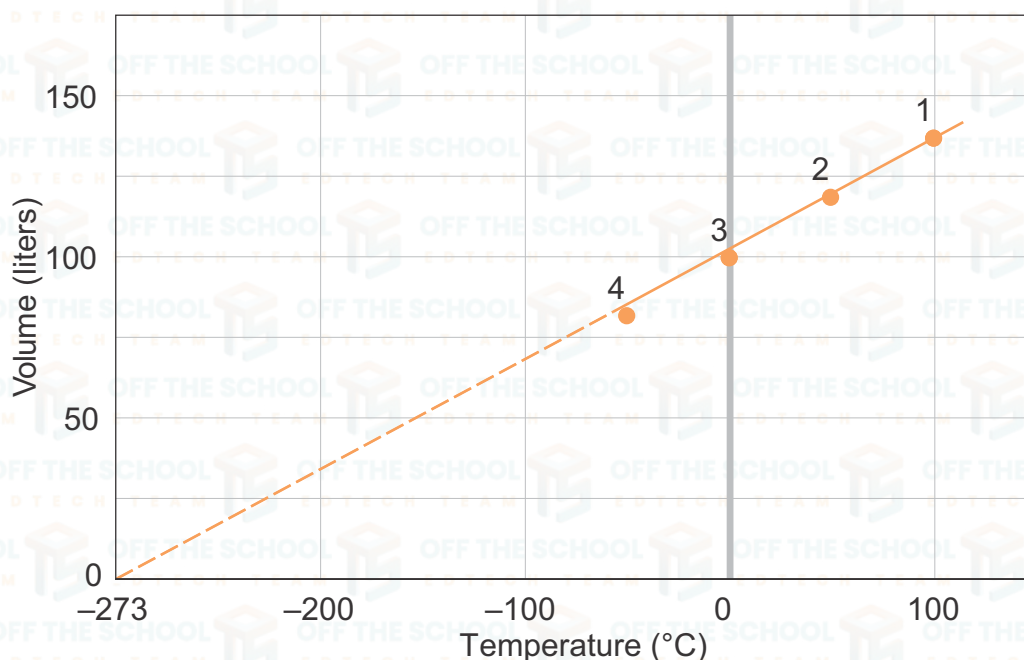
$$T(K) = [t(^{\circ}C) + 273] K$$



Graphical representation:

- If we plot a graph between volume and absolute temperature, a straight line is obtained, but with an intercept.
- Therefore we extrapolate a graph on (-ve) x-axis which cuts the graph at -273.15°C
- Where for a real gas is impossible to maintain its state therefore we change the scale where it is impossible to a real gas to reach -273.15°C is considered as Absolute Zero or 0 K.
- According to Charles, at 0 K volume of gases should be zero.
- Absolute Zero is defined as: **The temperature at which the volume of an ideal gas becomes zero.**
- For real gases, the concept of absolute zero does not applied.
- The inter-conversion between $^{\circ}\text{C}$ & K is as:

$$\text{K} = ^{\circ}\text{C} + 273.15$$

**AVOGADRO'S LAW:****Introduced By:**

Amedo Avogadro (1811).

Study of:

Study of the relationship between volume & number of moles of gases at constant pressure & temperature.

Statement:

Under similar conditions of temperature & pressure, equal volumes of all gases contain an equal number of moles.

OR

The volume of a gas is directly proportional to the number of moles of gas if temperature & pressure is kept constant.

$V \propto n$ (at constant pressure & temperature).

$$\frac{V}{n} = \text{Constant}$$

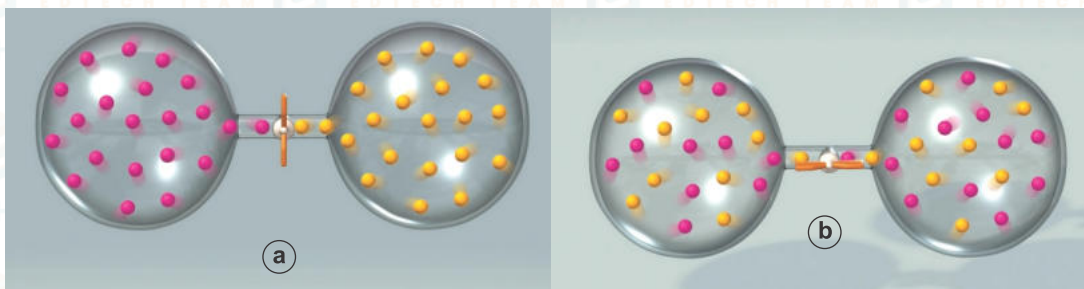


The addition of carbon dioxide to the balloon increases the number of gas molecules, which increases the volume.



DIFFUSION:

"Spreading of a substance is diffusion or the movement of a substance from their higher concentration to lower concentration."

**EFFUSION:**

"When a gas escapes through a pin-hole into a region of low pressure or vacuum, the process is called Effusion."

GRAHAM'S LAW:**Introduced by:**

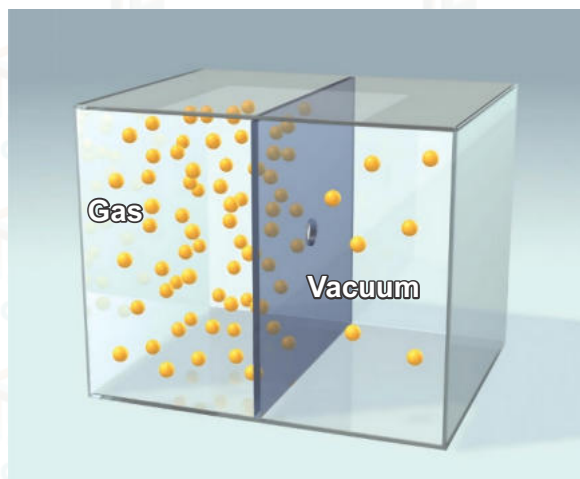
Thomas Graham (1831) for the relationship between diffusion rate & density of gas.

Modified by:

Thomas Graham (1848) for the relationship between diffusion rate & molar mass of gas.

Statement:

"At constant temperature & pressure, the rate of effusion or diffusion of a gas is inversely proportional to the square root of its density or molar mass."



▲ Effusion is the escape of a gas into a vacuum.

If;

- ✓ 'r' is the rate of diffusion or effusion.
- ✓ 'd' is the density of a gas.
- ✓ 'M' is the molar mass.
- ✓ 't' is the time of diffusion or effusion.
- ✓ 'V' is the volume of the gas, then according to Graham's law:

$$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}} = \frac{t_2}{t_1} = \frac{V_1}{V_2}$$

APPLICATIONS OF GRAHAM'S LAW:

- Densities & molar masses of unknown gases can be determined.
- The effect of toxic gases can be reduced by diffusing them in air.
- Various isotopes of gases can be separated due to mass distinction.

REASONS FROM PAST PAPER:**1. The rate of diffusion of CO_2 and C_3H_8 (Propane) are the same. (2016, 08)**

According to Graham's law, the rate of diffusion is inversely proportional to the square root of the molar masses. Since the molar masses of CO_2 and C_3H_8 are nearly equal, their rates of diffusion are also similar.

2. Process of diffusion occurs most rapidly in gases, less in liquids and very slowly in solids. (2011,05,00)

Graham's law states that the rate of diffusion is directly proportional to the average speed of the particles, which is highest in gases, lower in liquids, and the lowest in solids due to differences in intermolecular forces and freedom of movement.

3. 100 cm^3 of O_2 and 100 cm^3 of NH_3 contain the same number of molecules at STP. (2008)

At STP, equal volumes of gases contain an equal number of molecules. This is explained by Avogadro's law, stating that equal volumes of gases at the same temperature and pressure contain the same number of molecules, regardless of their chemical identity.

DALTON'S LAW OF PARTIAL PRESSURE:

Introduction:

An English Chemist John Dalton (1766–1844) gave law of partial pressure in 1801.

Statement: According to this law:

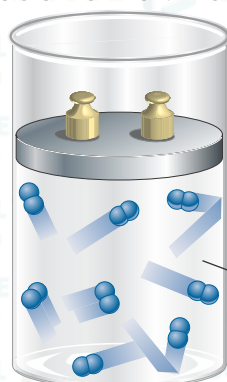
"The total pressure exerted by a gaseous mixture is equal to the sum of partial pressures of each gas present in the mixture."

$$\text{Total Pressure (P}_t\text{)} = P_1 + P_2$$

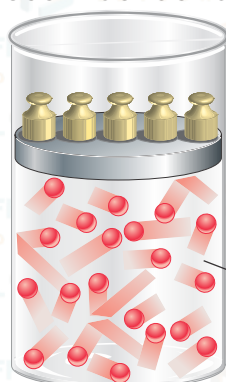
Whereas P_1 & P_2 are the partial pressures of individual gases.

CONDITIONS:

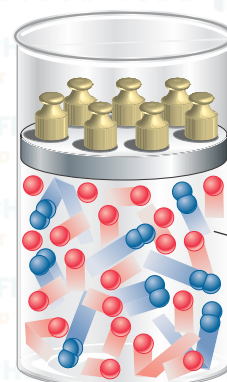
- The law is only obeyed, if the component gases do not chemically react with each other. Moreover, the gases must behave ideally.



(a) 5.0 L at 20 °C



(b) 5.0 L at 20 °C



(c) 5.0 L at 20 °C

- The partial pressure is the pressure exerted by one component of the gaseous mixture.

Example:

Air is a mixture of non-reacting gases, Percentage of each gas in the air is:

- (1) N_2 (78.08%), O_2 (20%), Ar (0.93%), CO_2 (0.03%).
- (2) Traces of Ne, He, Kr, H_2 .
- (3) Various amounts of water vapours.
- (4) Pollutant (which can pollute the atmosphere) gases such as oxides of sulphur and nitrogen.

The total pressure exerted by the air will be the sum of partial pressures of each gas.

MATHEMATICAL FORM:

The pressure exerted by each gas when it is present alone in a mixture is called partial pressure.

$$P_1 = P_A + P_B + P_C$$

$$PV = nRT$$

By using ideal gas equation,

$$P = \frac{n.R.T}{V}$$

Thus, in a mixture of non-reacting gases, molecules of each gas behave independently. Therefore, molecules of each gas have an equal opportunity to collide with the wall of the container & exert pressure.

Therefore, for each gas ideal gas equation can be written as;

For gas A,

$$P_A = n_A.R.T/V$$

For gas B,

$$P_B = n_B.R.T/V$$

For gas C,

$$P_C = n_C.R.T/V$$

Therefore, Total pressure of a gaseous mixture can be obtained by adding partial pressure of each gas as,

$$P_t = \frac{n_A.R.T}{V} + \frac{n_B.R.T}{V} + \frac{n_C.R.T}{V}$$

$$P_t = n_A + n_B + n_C . R . T / V$$

$$P_t = \frac{n_t.R.T}{V}$$

Dividing pressure of individual gases by total pressure.

$$P_t = \frac{\frac{n_A.R.T}{V}}{\frac{n_t.R.T}{V}}$$

$$\frac{P_A}{P_t} = \frac{n_A}{n_t}$$

As, $\frac{n_A}{n_t}$ expressed as X_A (mole fraction).

Therefore,

$$\frac{P_A}{P_t} = X_A$$

$$P_A = X_A P_t$$

This equation indicates that the Partial pressure of any gas present in the mixture is equal to the product of the mole fraction & total pressure of the gaseous mixture.

APPLICATION OF DALTON'S LAW OF PARTIAL PRESSURE:

1) Pressure of Gas Collected Over Water:

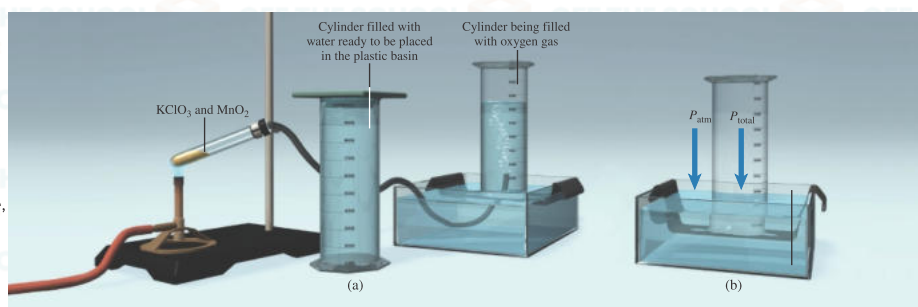
When gas is collected over water by downwards displacement of water in a gas jar. By applying Dalton's law of partial pressure, we can calculate the partial pressure of dry gas.

$$P_t = P_{\text{dry}} + P_{\text{watervapors}}$$

$$P_{\text{dry}} = P_t - P_{\text{watervapors}}$$

2) Maintenance of Oxygen Pressure at High Altitudes:

• The pressure at high altitudes is lower than sea level due to a decrease in the number of molecules of gas.



- (a) Apparatus for measuring the amount of gas produced in a chemical reaction.
 (b) When the water levels inside and outside the collection vessel are the same, the pressure inside the vessel is equal to atmospheric pressure.

• Respiration depends upon the partial pressure of oxygen in the air (159 torr) & in the lungs (116 torr) & at high altitudes there exist problems in breathing.

3) Maintenance of Oxygen Pressure for Deep Sea Divers:

- The pressure under the sea is higher than the sea level due to an increase in the partial pressure of oxygen gas.
- For example, at depth of 40 meters, pressure increases four times. This also causes breathing problems
- Therefore, sea divers use SCUBA (self-contained underwater breathing apparatus) with 96% He & 4% O₂.

VANDER WAAL'S EQUATION:

Introduced by:

J.D. Vander Waal (1873)

Purpose:

To modify two, conflicting postulates of kinetic molecular theory. The Equation that gives a mathematical solution by the correction in molecular volume & Intermolecular forces.

IDEAL GAS EQUATION:

The ideal gas equations can be expressed as:

$$PV = nRT$$

CONFLICTING POSTULATES:

- The actual volume of gas molecules is negligibly small compared to the total space of a container.
- Gas molecules neither possess attractive nor repulsive forces.

VOLUME CORRECTION:

- When pressure is applied to gas, the molecules come closer to each other.
- By continuous increase in pressure, a point is reached when molecules can't be further compressed due to repulsive forces.
- This indicates although the volume of gas molecules is very small but not negligible,
- Vander Waal calculated the actual volume of gas as:

$$V = V_{\text{vessel}} - nb$$

where,

$V = V_{\text{ideal}}$ = free volume

V_{vessel} = volume of a vessel in which gas molecules are present.

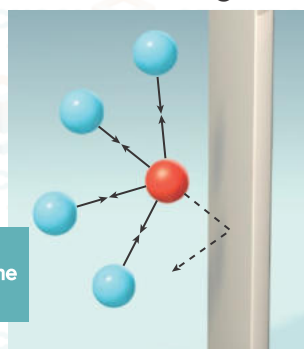
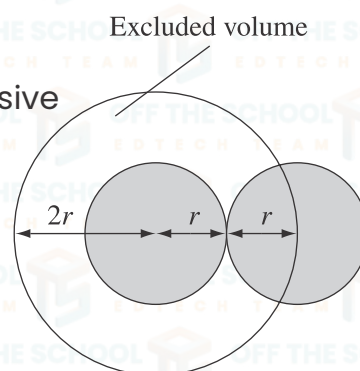
n = number of moles,

b = excluded volume of gas molecules per mole in highly compressed gaseous state (theoretical volume)

- The value of "b" depends upon the size of molecules of gas, "b" is not an actual volume of a molecule but it is 4 times that of the actual volume of 1 mole of a gas.

PRESSURE CORRECTION:

- Vander Waal also corrected the pressure produced by the molecules of real gas.
- On increasing pressure, molecules of gas come close to each other & intermolecular attractions increases.
- Consider a molecule "A" in the interior of gas which is surrounded by another



The effect of Intermolecular attractions on the pressure exerted by a gas.



molecule "B". The resultant attractive forces around "A" is zero.

- The molecule "A" when approaches to strike the wall of the container, experienced net inward pull due to attractive forces on B
- This attractive pull produces some reduction in pressure on the walls of the container which is given as;

$$P_{\text{observed}} = P_{\text{ideal}} - P_{\text{less}}$$

$$P_{\text{ideal}} = P + P_{\text{less}} \dots\dots\dots (i)$$

- The pressure drop is directly proportional to the number of particles per unit volume.

$$P_{\text{less}} = P_{\text{ideal}} - P_{\text{less}}$$

$$P_{\text{ideal}} \propto [A] \cdot [B]$$

$$P_{\text{less}} \propto \frac{n}{V} \cdot \frac{n}{V}$$

$$P_{\text{less}} \propto \frac{n^2}{V^2}$$

$$P_{\text{less}} = \frac{an^2}{V^2}$$

Insert the value of P less in equation (i):

$$P_{\text{ideal}} = P + \frac{an^2}{V^2}$$

Put the value of volume correction & pressure correction in the ideal gas equation.

$$PV = nRT$$

$$\left(\underbrace{P + \frac{an^2}{V^2}}_{\text{corrected pressure term}} \right) \left(\underbrace{V - nb}_{\text{corrected volume term}} \right) = nRT$$

experimentally measured pressure points to P
container volume points to V
corrected pressure term points to $P + \frac{an^2}{V^2}$
corrected volume term points to $V - nb$

This is known as the Vander Waal equation.

CHANGE IN THE STATE OF A GAS

DEFINITIONS:

- ✓ **Liquefaction:** The physical process in which gases are converted into liquids.
- ✓ **Critical temperature:** The highest temperature is where gas can exist as a liquid.
- ✓ **Critical pressure:** The pressure that is required at critical temperature to convert gas into liquid.
- ✓ **Critical Volume:** At critical temperature & pressure, the volume of 1 mole of a gas.

CAUSES OF LIQUEFACTION OF GASES:

- Gas molecules at high pressure & low temperature come closer & attractive forces are pronounced.
- If the temperature is further lowered below the critical temperature of a gas, the attractive forces hold the molecules together to form a liquid.
- Each gas has a specific critical temperature & pressure, by maintaining these parameters a gas can be liquefied.

Methods of liquefaction

Faraday's method

Claude's method

Linde's method

METHODS OF LIQUEFACTION OF GASES:

JOULE-THOMSON EFFECT KELVIN-JOULE EFFECT OR JOULE-KELVIN EFFECT:

When highly compressed gas is allowed to expand from a region of higher pressure to lower pressure through a nozzle, a fall in temperature is observed.



**LINDE'S METHOD:**

Introduced by Hampson Linde.

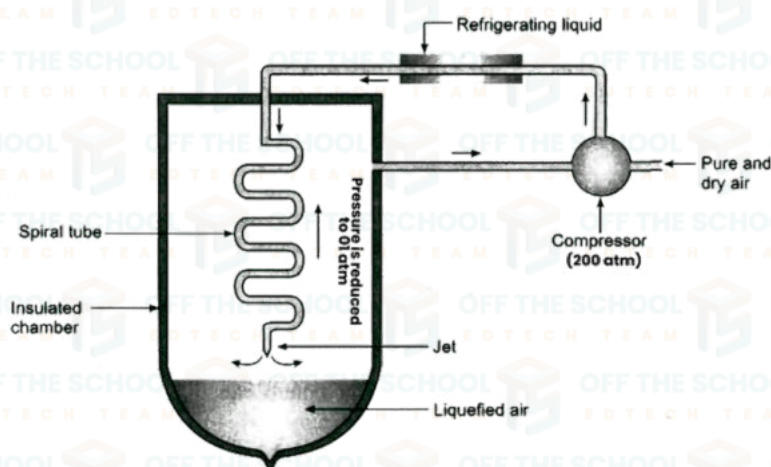
When a gas is allowed to expand adiabatically from a region of high pressure to a region of extremely low pressure, the liquefaction takes place.

Reason:

An obvious reason for the cooling is that when the gas expands from a region of high pressure to the region of low pressure, the intermolecular distance increases. Some work is to be done for increasing intermolecular distances. This is done on the cost of Kinetic energy of the gas. Thus, the kinetic energy of the gas decreases and therefore the temperature falls.

Process:

1. First of all, gas is compressed, to 200 atm by the compressor.
2. Compressed air is allowed to pass through pipes containing water. Here moisture present in the air is condensed & water is removed.
3. The heat of compressed air is absorbed by water, & dry air is passed through a spiral copper coil having a jet at another end.
4. On passage of dry gas due to expansion, pressure is reduced to 1 atm causing a decrease in temperature.
5. As a result, cool air is formed which moves up & cools the upcoming compressed air.
6. Uncondensed gas is sent to a compression pump for recycling.
7. The process of cooling, compression & expansion is repeated several times.
8. Liquid gas (or air) is collected below & removed.

**PLASMA:(non-evaluative):**

Identified by W. Crookes (1879) while working on the discharge tube experiment named Radiant Matter. Irving Langmuir (1920) used the term plasma for it. This is the 4th state of matter most of the universe consists of the matter in plasma state.

- A plasma is an extremely ionized gas.
- A plasma is a very good conductor of electricity and is affected by magnetic fields.
- Plasma, like gases have an indefinite shape and an indefinite volume.



Occurrence:

99% of a visible universe is composed of plasma. For example in Sun, stars, nebulae & ionosphere but rarely on earth such as lightning bolts, flames, welding arcs, auroras & fluorescent.

Production:

When gas is heated at a high temperature $10^4 - 10^5$ K, most of the molecules are ionized with the formation of clouds of electrons along with the presence of un-ionized atoms & molecules.

Properties:

- Only exist at high-temperature or low-temperature vacuums.
- Natural plasma is extremely hot & does not break down or react rapidly.
- It can vaporize matter.
- Plasma contains large numbers of oppositely charged ions but it is overall neutral.
- It responds to electric & magnetic fields.
- Plasma doesn't possess a particular shape or volume.
- Plasma shows a characteristic glow in the discharge tube depending upon the nature of gas.

Uses:

- When the light of a fluorescent bulb or neon sign is turned on, gas is excited & creates glowing plasma which lightens the surrounding.
- Used in television & computer chips.
- Used in rocket propulsion.
- Used in cleaning the environment.
- Used in destroying biological hazards & healing wounds.
- Used in the processing of semiconductors.
- Used in sterilization of some medical products.
- Used in lasers, lamps, pulsed power switches & diamond-coated films.

